NVTool User Guide

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1 Overview

1.1 NVTool Introduction

1.1.1 Toolkit

NVTool is unisoc NV2.X project's toolkit，It include below three applications：

Name	Descriptions
NVTool.exe	GUI Tool
CmdNV.exe	Window command line application
LinuxCmdNV	Linux command line application

table 1.1 Toolkit

1.1.2 Tool Introduction

NVTool is unisoc NV2.X project's GUI tool，support below functions：

Function	Descriptions
Edit NV	Edit and modify nv data
Generate NV Image	Generate NV image file
Read nv data from phone	Read nv data from phone via usb or uart
Wrtie nv data to phone	Wrtie nv data to phone via usb or uart

table 1.2 NVool function

1.2 Environment

1.2.1 Hardware environment

Hardware	basic requirements
PC Memery	4G and above
PC hard disk	4G and above free space
PC USB Port	At least one USB port
Cable	Uart or usb cable

table 1.3 Hardware Configuration

1.2.2 Software environment

Software	basic requirements
OS	Windows 7 and above
USB driver	2.0.0.131 and above

table 1.4 Software Configuration

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2 Quick Start

This chapter describes how to use NVTool to do NV by quick steps

2.1 Quick Start

Click NVTool.exe to launch, GUI is as below ("R1.19.2501" is tool's demo version,The version number will be continuously upgraded, which is based on the actual tool interface.)

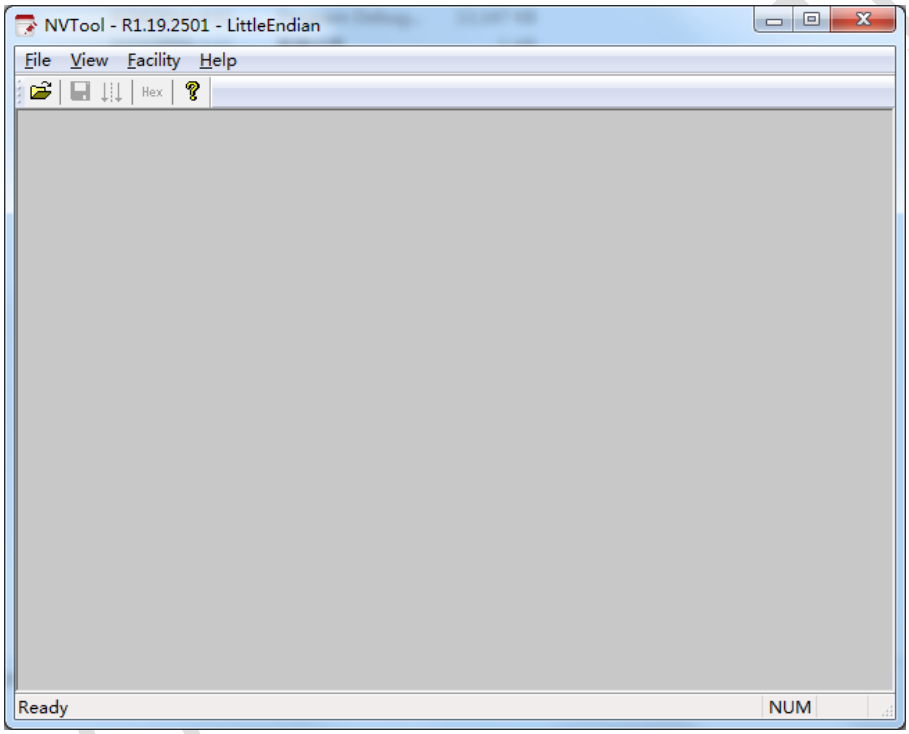


Photo 2.1 Main UI

Step1 : Open NV project file

You can open the nv project in the following ways :

- Select [File\Open Project] menu
- Ctrl+O Shortcut
- Click  icon on toolbar
- [File] There is a list of recently opened projects under the menu item , Directly select one of the recent NV projects.

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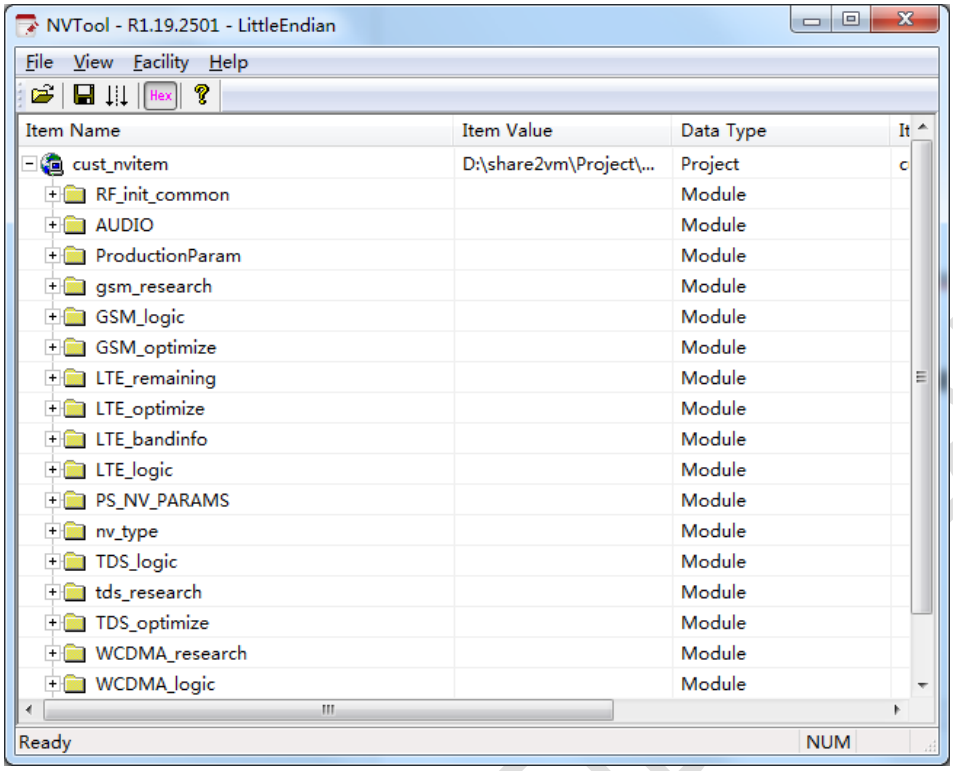


Photo 2.2 Loaded NV Project

Step2 : Edit NV parameter

Double-click the value you want to modify or select the value you want to modify. Press the Enter key. An edit box will appear and you can enter the new value.

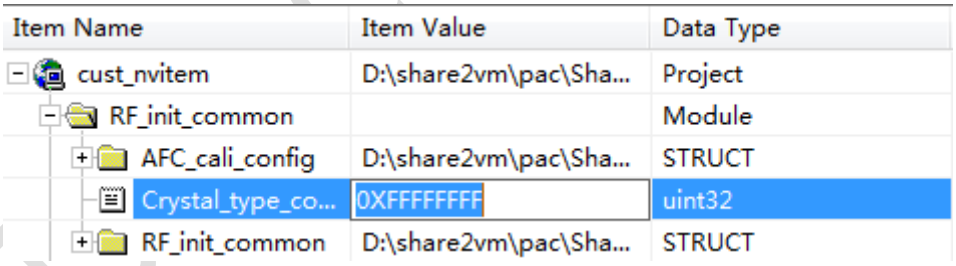


Photo 2.3 Editor NV Parameter

Step3 : Generate NV Image

Select [File\Save Image] menu to generate a new nv image file，The new NV image file is saved in the same directory of the nv project.

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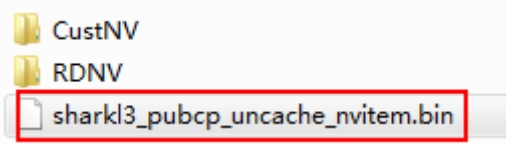


Photo 2.4 Generated NV Image file

Step4 : Save NV project

Select [File\Save Project] menu or Click icon on the toolbar, then can generate nv image file and save nv project.

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3 Operation Guide

3.1 Toolbar



Phote 3.1 Toolbar

Icon	Descriptions
	Open project
	Save project
	Save nv parameter to phone (it is same as [File \Save To Phone (Normal mode)] menu
	Decimal or hexadecimal display nv parameter switch,defult is hexadecimal
	Help

table 3.1 Toolbar

3.2 Menu

3.2.1 File menu

[File] menus as below:

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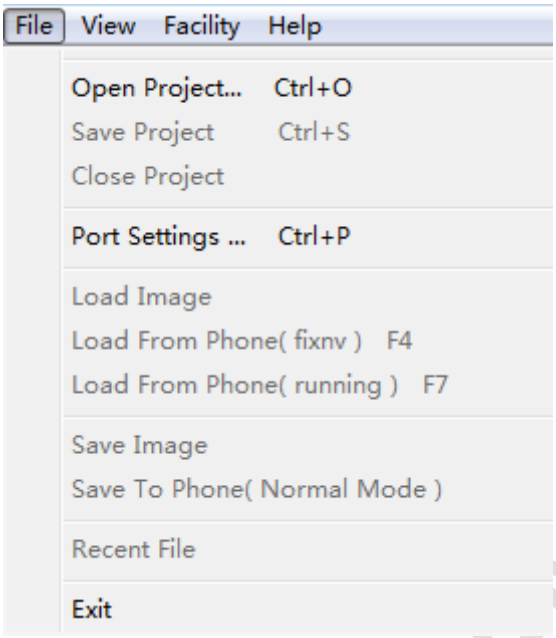


Photo 3.2 File Menu

Menu	Descriptions
Open Project	Open nv project or nv xml file
Save Project	Save nv project and save nv image file
Close Project	Close nv project
Port Settings	Port setting, using for read nv from phone or write nv to phone
Load Image	Load nv image to nv project
Load From Phone (fixnv)	Load fixnv from phone, phone, The device needs to enter calibration mode, port is u2s diag device
Load From Phone (running)	Load running nv from phone, default is disable, If you need to use this function, you can modify below config in NVTool.ini file [GUI] EnableLoadRunningNvMenu = 1
Save Image	Save nv image file
Save to Phone (Normal Mode)	Save nv items to phone
Exit	Exit application

table 3.2 File Menu

3.2.1.1 Load Image

This function can parse the NV image file and import the NV parameters of the selected NV

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image into the open NV project.

3.2.1.2 Port Settings

[File\Port Settings] menu and Ctrl+P shortcut can open port settings windows, this port setting just for read nv parameters from phone or write nv parameters to phone. Please select this function menu before connected the device.

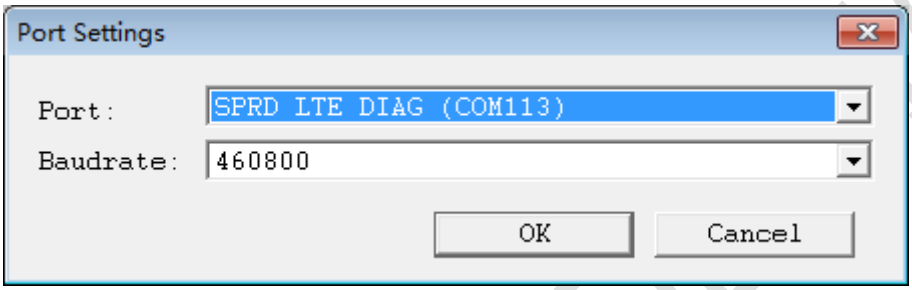


Photo 3.3 Port Settings

Item	Descriptions
Port	Uart or usb port
Baudrate	Baudrate, just for uart connect

table 3.3 Port Setting

3.2.1.3 Load From Phone (fixnv)

After loading the NV project, you can use this menu function to read the device's fixnv in calibration mode. Additional tools are required to bring the device into calibration mode.

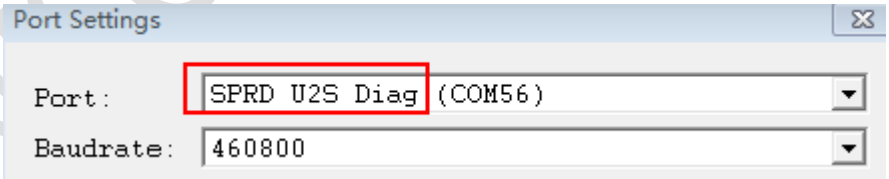


Photo 3.4 Calibration devices

3.2.1.4 Load From Phone (runningnv)

This feature is turned off by default , This feature can be turned on by modifying the configuration. After loading the NV project, you can use this menu function to read the runningnv of the device in the boot/calibration mode. If you read in the boot mode, you need

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to ensure that the Modem to PC function is enabled in the engineering mode of the device.

3.2.1.5 Save To Phone (Normal Mode)

After loading the NV project, you can use this menu function to write nv parameters to the runningnv of the device in the boot/calibration mode.

3.2.2 Facility Menu

[Facility] Menu as below:

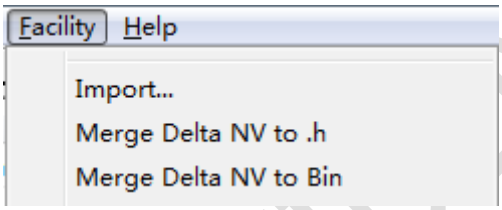


Photo 3.5 Facility Menu

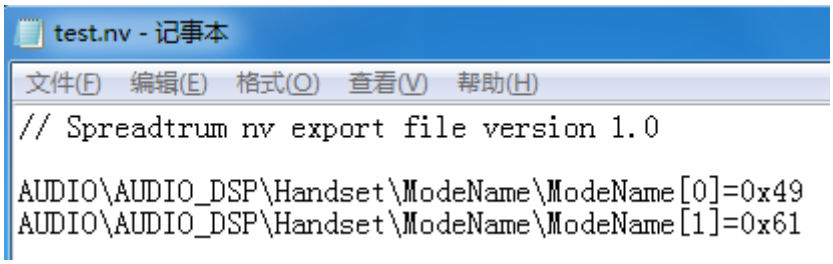
Item	Descriptions
Import	Import nv file
Merge Delta NV to .h	Generate nv_config.h file under the nv project directory
Merge Delta NV to Bin	Generate delta_nv.bin file under the delta nv directory

table 3.4 Facility Menu

3.2.2.1 Import

This feature supports importing NV file content into a loaded NV project , The imported difference item will displayed by red color. NV file is generally exported from NVTool, Audiotest, NVAide, etc. or manually generated .nv files. The left side of the equal sign is the NV path, The right side of the equal sign is the NV value. Format is as below:

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Poto 3.6 NV file format

Demo after import nv file:

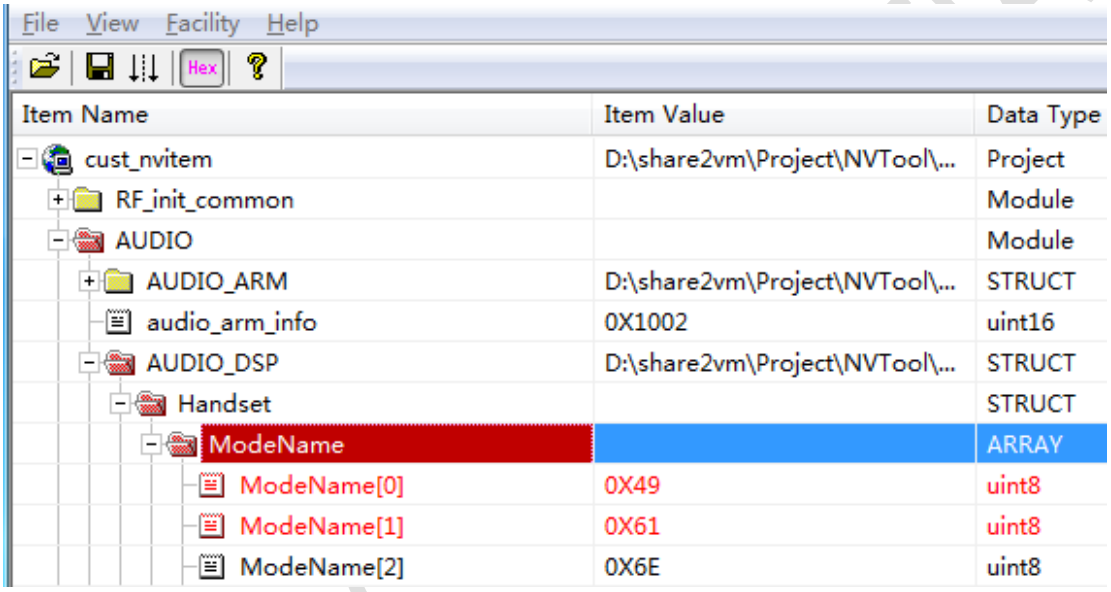


Photo 3.7 Import NV File

Explain：

If the same nv item but difference value exists in the imported delta nv file, only the last one is imported into the nv project。

e.g. If the following two differential nv appear, the value that is finally imported into the project is actually 0x62

AUDIO\AUDIO_DSP\Handset\ModeName\ModeName[1]=0x61

AUDIO\AUDIO_DSP\Handset\ModeName\ModeName[1]=0x62

3.2.2.2 Merge Delta NV to Bin

This function is used to generate delta nv image file.

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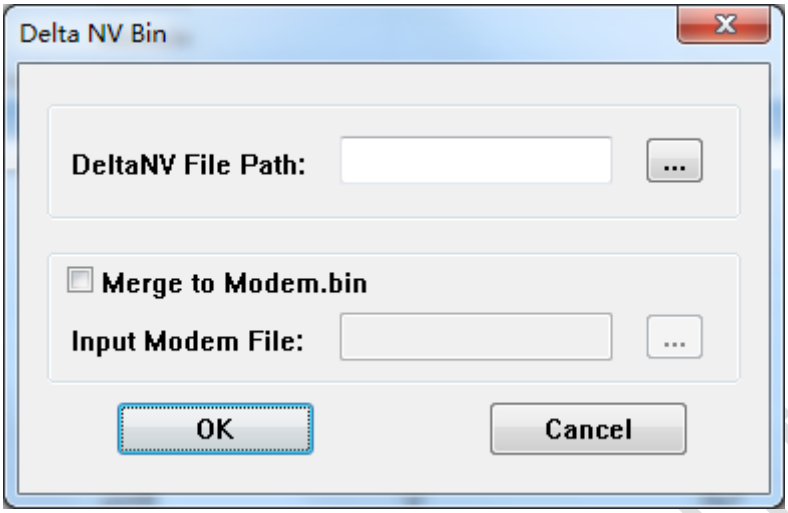


Photo 3.8 DeltaNV Bin Settings

Item	Descriptions
Delta NV File Path	The directory of delta nv file (required)
Merge to Modem.bin	1. If not selected, the delta_nv.bin file will be generated in the selected DeltaNV file path 2. If selected，must select one modem file at the same time，Then will merge the generated delta_nv.bin file into the input modem file.
Input Modem File	Input modem file
OK	Click it and will generate file

table 3.5 Delta NV Bin Settings

3.2.3 Help Menu

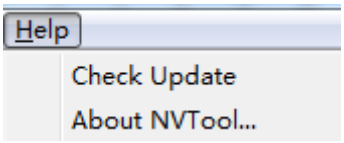


Photo 3.9 Help Menu

Item	Descriptions
Check Update	Check update NVtool
About NVTool	About NVTool infomation

table 3.6 Help Menu

3.2.3.1 Check Update

Checking the version function requires the PC to access the network ,This function can manually check whether

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the NVTool has an updated version, if there is an updated version, will show below upgrade UI, User can click Upgrade Now button to upgrade NVTool. If user do not select "Do not prompt new version any more" option, then The upgrade will also be automatically checked when launched the NVTool.

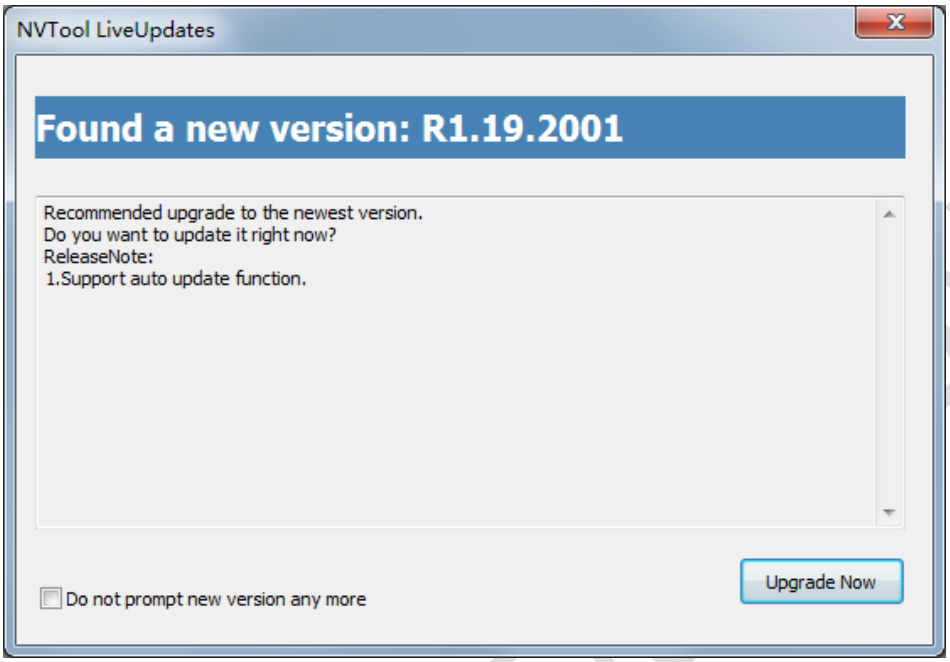


Photo 3.10 Upgrade

If not new version NVTool, will show below information.

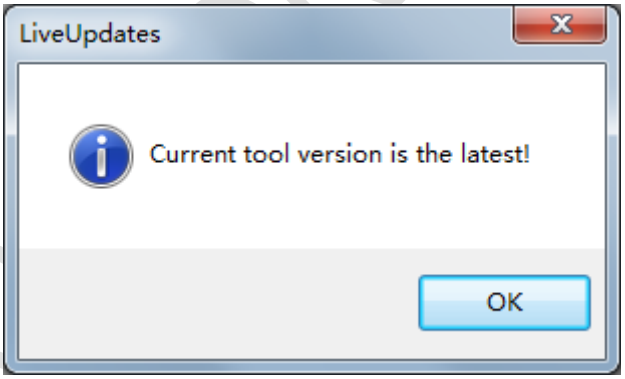


Photo 3.11 No new version

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4 Command Line Application

4.1 Overview

CmdNV.exe and LinuxCmdNV tool is Windows and Linux command line application，The two sets of command line compilation tools have the same function and usage. This article only uses the CmdNV application as an example，The command line compilation tool supports the following features：

- Build xml and generate nv bin and header file
- Generate delta nv functions
- Import NV file and Generate nv image file

4.2 Operation Guide

4.2.1 Parameter formate

Import –h or-help parameter, can read command line application’s help.

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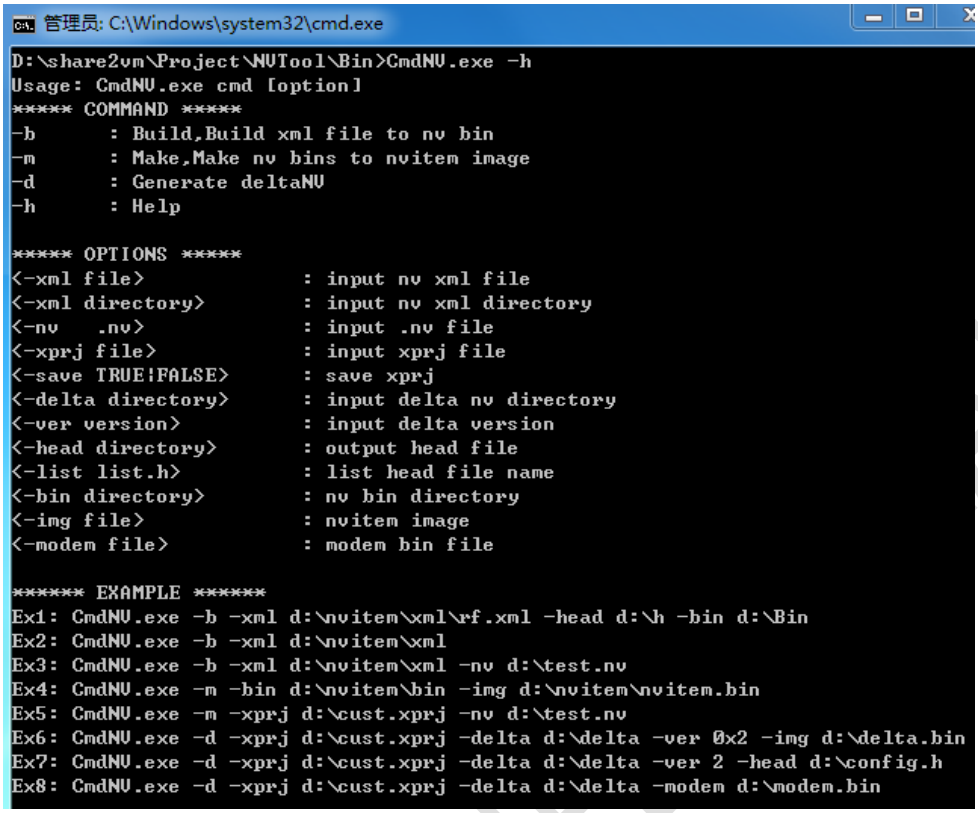


Photo 4.1 Command line parameter

4.2.2 Parameter explain

4.2.2.1 Build xml file

Compile the generated header file and bin file to the specified directory，This function is only used when compiling the modem module. Customers generally do not need this function.

e.g. CmdNV.exe -b -xml d:\nvitem\xml\rf.xml -head d:\h -bin d:\Bin

4.2.2.2 Build nv project and generate NV Image file

Ex1: build nv project and generate nv image file

CmdNV.exe -m -xprj d:\nvitem\RDNV\rd_nvitem.xprj

Ex2: import nv file and generate nv image file, it will import nv parameter to the nv project.

CmdNV.exe -m -xprj d:\nvitem\RDNV\rd_nvitem.xprj -nv d:\test.nv

Ex3: Import nv directory and generate nv image file.

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CmdNV.exe -m -xprj d:\nvitem\RDNV\rd_nvitem.xprj -nv d:\testnv\

4.2.2.3 Build delta nv file

Ex1: Build and generate delta bin file

CmdNV.exe -d -xprj d:\nvitem\RDNV\rd_nvitem.xprj -delta d:\delta -ver 0x2 -img d:\delta.bin

-delta : input delta nv directory，-img : output delta bin file，-ver : Need to enter decimal or hexadecimal content parameters starting with 0x. default not setting.

Ex2: Build and generate delta header file

CmdNV.exe -d -xprj d:\nvitem\RDNV\rd_nvitem.xprj -delta d:\delta -ver 2 -head d:\config.h

-delta : input delta nv directory，-head : output delta head file，-ver : Need to enter decimal or hexadecimal content parameters starting with 0x. default not setting.

Ex3: Generate delta bin file and merge to modem bin file

CmdNV.exe -d -xprj d:\nvitem\RDNV\rd_nvitem.xprj -delta d:\delta -modem d:\modem.bin

-delta : input delta nv directory，-modem : input /output modem file

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Revision history

Version	Date	Author	Description
V1.0	2019-06-14	UNISOC	First version.

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